

DAY-9 — DNS Basics

CyberNotes • Detailed Beginner Reference (Improved Edition)

Learning goal: Build a practical, working understanding of **dns basics**, connect the concept to real cybersecurity work, and practice it safely in a local lab.

Study rule: Understand a command or concept before copying it. Security tools are powerful because they interact with real systems — your practice environment should always be intentionally controlled and authorized.

1. Core Concepts

Core idea: DNS Basics is a building block for practical cybersecurity. This section explains what each term actually means, where you'll encounter it, and how it connects to the bigger picture.

Why it matters: Security work depends on understanding normal system and network behavior before you can recognize abnormal behavior. This topic connects directly to Linux, networking, web security, monitoring, and troubleshooting — the threads running through this whole 20-day series.

Vocabulary covered: DNS purpose, records, resolution, recursive queries, caching, TTL, dig, nslookup.

Reference	What to remember
<code>dig example.com</code>	DNS lookup
<code>dig A example.com</code>	A record
<code>dig MX example.com</code>	mail records
<code>dig +trace</code>	resolution path
<code>nslookup</code>	basic DNS query
<code>resolvectl status</code>	resolver state

2. Concept Explanations

Each term below is explained in plain language, with a note on why it matters for security work. Read these before running any commands.

1. DNS purpose

DNS (Domain Name System) translates human-friendly domain names (like `example.com`) into the IP addresses computers actually use to connect. It's often called the "phone book of the internet," and because so much depends on it, DNS is also a common attack surface — for things like domain hijacking, cache poisoning, and using DNS queries to exfiltrate data.

2. records

DNS stores different types of records for different purposes: A (IPv4 address), AAAA (IPv6 address), MX (mail server), CNAME (alias to another name), TXT (arbitrary text, often used for domain verification or SPF/DKIM email security). Knowing which record type you need tells you which `dig` query to run.`

3. resolution

Resolution is the process of turning a domain name into an IP address by querying DNS servers, ultimately starting from the root servers and working down through top-level domain servers to the domain's own authoritative server. Understanding this chain helps you figure out exactly where a resolution failure is happening.

4. recursive queries

A recursive query asks a resolver (often your ISP's or a public one like 8.8.8.8) to do all the work of finding the final answer, following the chain on your behalf and returning just the result. This is what your computer does by default every time it looks up a name.

5. caching

DNS caches store recently resolved names for a period of time to speed up future lookups and reduce load on authoritative servers. Caching is efficient but means changes to DNS records don't propagate instantly — old ("stale") answers can keep being served until the cache expires.

6. TTL

TTL (Time To Live) is the number of seconds a DNS record is allowed to be cached before it must be re-fetched, set by the domain owner. A low TTL means changes propagate faster but generates more query traffic; a high TTL is more efficient but slower to update — this matters a lot when migrating a service to a new server.

7. dig

``dig`` is the standard tool for querying DNS directly and inspecting the raw response, including which server answered and how long the record has left to live. ``dig +trace`` is especially useful for security and troubleshooting because it shows the entire resolution path, hop by hop, rather than just the final answer.

8. nslookup

``nslookup`` is an older, simpler DNS query tool, still common on Windows and useful for quick checks even though ``dig`` offers more detail on Linux. Either tool answers the same basic question — what does this name resolve to right now — just with different output formats.

9. common issues

Common DNS issues include misconfigured records, expired domain registrations, propagation delays after a change, and resolver misconfiguration on the client side. ``resolvectl status`` shows which resolver your machine is actually using, which is often the first thing to check when DNS seems to behave inconsistently.

3. Safe Practical Lab

Step 1. Define the lab target and the exact question you are trying to answer.

Step 2. Run the smallest safe observation or command that can answer that question.

Step 3. Read the output carefully; record important fields, errors, timestamps, and assumptions.

Step 4. Change one variable at a time and compare the result.

Step 5. Write a short conclusion and clean up temporary files or services.

Example lab sequence:

```
dig example.com
dig A example.com
dig MX example.com
```

4. Common Mistakes & Troubleshooting

- Running a command without checking its target, current directory, or privileges.
- Confusing a tool's output with proof of a vulnerability or compromise — output is evidence, not a conclusion.
- Ignoring error messages instead of reading the exact reason for failure.
- Changing several settings at once, making it hard to tell what actually fixed or broke the lab.
- Saving real credentials, tokens, or personal data in practice notes or public repositories.
- Testing a public target just because it's reachable — reachability is never the same as authorization.

Troubleshooting checklist:

- What system am I on?
- What exact command/request did I run?
- What did I expect?
- What happened instead?
- What does the error message say?
- Can I reproduce it in a clean local lab?
- What changed since the last successful attempt?

5. Deep-Dive Questions

Answer these in your own words after finishing the lab — they're designed to check real understanding, not memorization.

Q1. What problem does DNS purpose solve, in your own words?

Notes: _____

Q2. What is the difference between normal behavior and a suspicious observation in this topic?

Notes: _____

Q3. Which command, log, packet, file, or response would you inspect first, and why?

Notes: _____

Q4. What could cause a misleading result or false positive here?

Notes: _____

Q5. How would you document your observation so another learner could reproduce it safely?

Notes: _____

Q6. Which security control reduces the risk associated with this topic?

Notes: _____

6. Revision Checklist

- I can explain dns basics in my own words.
- I can identify the main terms without looking at the notes.
- I can perform the example in a controlled lab.
- I can interpret the output instead of only copying it.
- I know at least one defensive or troubleshooting use for the concept.
- I can document the result clearly and safely.

7. Safety & Scope

Use these commands and tools only on systems you own, a local virtual machine, a CTF, an intentionally vulnerable application, or a target for which you have explicit written permission.

Keep a written scope: target, allowed actions, time window, and stop conditions. Never scan or test random public systems.

When practicing, save your notes and outputs so you can explain what you observed rather than blindly running commands.

Next step: After Day-9, review your notes and connect this topic to the next subject in the series. The goal is not to memorize commands — it's to understand systems well enough to reason about security.

8. Quick Reference & Personal Notes

Topic: Day-9 — DNS Basics

Three things I learned:

1. _____
1. _____
1. _____

Commands / concepts to memorize:

- `dig example.com` — DNS lookup
- `dig A example.com` — A record
- `dig MX example.com` — mail records
- `dig +trace` — resolution path
- `nslookup` — basic DNS query
- `resolvectl status` — resolver state

My lab result

What I tested: _____

What I observed: _____

What I would investigate next: _____

Final reminder: A strong cybersecurity learner can explain both the offensive concept and the defensive reason for learning it. Keep practice authorized, reproducible, and documented.